

Critical Perspectives

Population Modeling in Metal Risk Assessment: Extrapolation of Toxicity Tests to the Population Level

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Abstract: Population models can be a useful tool for ecological risk assessment to increase ecological realism. In the present study, population models were used to extrapolate toxicity test results of four metals (Ag, Cu, Ni, Zn) to the population level. In total, three primary producers, five invertebrate species, and five fish species were covered. The ecological modeling-based laboratory to population effect extrapolation factor (ECOPEX factor), defined as the ratio of the predicted 10% effect concentration (EC10) at the population level and the observed EC10 for the laboratory toxicity test, ranged from 0.7 to 78.6, with a median of 2.8 ($n = 27$). Population modeling indicated clearly higher effect concentrations in most of the cases (ECOPEX factor > 1 in 14 out of 27 cases), but in some cases the opposite was observed (in three out of 27 cases). We identified five main contributors to the variability in ECOPEX factors: (1) uncertainty about the toxicity model, (2) uncertainty about the toxicity mechanism of the metal, (3) uncertainty caused by test design, (4) impact of environmental factors, and (5) impact of population endpoint chosen. Part of the uncertainty results from a lack of proper calibration data. Nonetheless, extrapolation with population models typically reduced the variability in EC10 values between tests. To explore the applicability of population models in a regulatory context, we included population extrapolations in a species sensitivity distribution for Cu, which increased the hazardous concentration for 5% of species by a factor 1.5 to 2. Furthermore, we applied a fish population model in a hypothetical Water Framework Directive case using monitored Zn concentrations. This article includes recommendations for further use of population models in (metal) risk assessment. *Environ Toxicol Chem* 2024;43:2308–2328. © 2024 SETAC

Keywords: Extrapolation; Metal risk assessment; Population model; Toxicity model

POPULATION MODELING AND ECOLOGICAL RISK ASSESSMENT

Traditional ecological risk assessment (ERA) is mainly based on testing effects on organisms in the laboratory. The effects of metals are well studied in a wide range of species. Metal risk assessment is, therefore, typically based on deriving a safe threshold with a species sensitivity distribution (SSD), sometimes supported with mesocosm studies (Belanger & Carr, 2019). For most species, data included in the SSD are individual-level endpoints (e.g., mortality, reproduction). However, environmental protection goals typically focus on the population and community levels, not on individual-level effects. Moreover, there is a wide range of test design variables that can influence

the outcome of toxicity tests (e.g., no-observed-effect concentration [NOEC] or 10% effect concentration [EC10]), such as exposure duration, life stage tested, physicochemistry of the test medium, and temperature regime (Stark et al., 2004). Despite this, current ERA approaches mostly ignore this variability, when all data points (NOEC or EC10 values) are compiled within an SSD approach to derive safe thresholds, even though data points are not necessarily comparable. The introduction of population models into the ERA framework (Forbes et al., 2016; Grimm & Thorbek, 2014) can help reduce this variability. They can predict effects on a consistently comparable population-level endpoint from a variety of available information. Population models can help predict population effects over an extended period of time, extrapolating observed effects at the individual level to a more ecologically relevant context. Especially the extrapolation to untested scenarios is a great asset for population models (Forbes et al., 2016).

Population modeling has advanced substantially in the last two decades (Accolla et al., 2021; Forbes et al., 2008;

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TABLE 1: Overview of European risk-assessment schemes and their population protection goals stated

Regulation	Relevant sections on environmental protection	Population-level protection goal
Plant protection products (PPP) EC 2009 Reg 1107/2009	The PPP regulation specifies that "member states shall ensure that the use of plant protection products does not have any long-term repercussions for the abundance and diversity of nontarget species." Elements listed under "no unacceptable impact" include decrease in biodiversity, impact on ecosystem functioning and functionality, and decrease in perceived aesthetic value (including disappearance of species and visible mortality of vertebrates).	The aquatic key drivers and their ecological entity to be protected as proposed by the EFSA PPR Panel: • Abundance and diversity of aquatic algae, aquatic vascular plants, aquatic invertebrates and aquatic vertebrates. • Aquatic vertebrates (fish): also individuals (in acute risk assessment to avoid visible mortality).
Regulation on the Evaluation, Authorisation and Restriction of Chemicals (REACH) EC 2006 Reg 1907/2006	The environmental hazard assessment shall consider the following: (1) the environment (including sediment), (2) terrestrial and (3) atmospheric compartments, and the potential effects that may occur (4) by means of food-chain accumulation and on the (5) microbial activity of sewage treatment systems. The approach is focused on the protection of the most sensitive species against long-term exposure.	No explicit mention of population protection goals.
Biocidal products regulation (BPR) EC 2012 Reg 528/2012	"No unacceptable effect ... on the environment having particular regard to contamination of surface waters (including estuarine and seawater), groundwater and drinking water, air and soil (and) its impact on nontarget organisms, (and) impact on biodiversity and the ecosystem." In addition, BPR considers "any adverse effects arising in any of the three environmental compartments—air, soil and water (including sediment)—and of the biota following the use of the biocidal product." It is assumed that ecosystem sensitivity depends on the most sensitive species groups and that protecting ecosystem structure protects community function.	No explicit mention of population protection goals.
Water Framework Directive (WFD) EC 2000 Dir 2000/60/EC	Under the WFD, member states should aim to achieve the objective of at least a "good ecological status" and a "good chemical status" by defining and implementing the necessary measures. The ecological status is defined as "an expression of the quality of the structure and functioning of aquatic ecosystems with surface waters." For specific substances, so-called priority pollutants, environmental quality standards are set, which are environmental safety thresholds that should not be exceeded. It is assumed that ecosystem sensitivity depends on the most sensitive species groups and that protecting ecosystem structure protects community function.	Protection goals for surface waters defined per biological quality element: • Phytoplankton (not for rivers), macrophytes, and phytobenthos: only community protection goals (e.g., composition) • Benthic invertebrates: taxonomic abundance • Fish: species abundance, presence disturbance-specific species, age structure

EFSA = European Food Safety Authority; PPR = Plant Protection Products and Their Residues.

population growth rate can be highly dependent on (assumed) initial abundance during population simulations and can be highly variable over the course of the simulated population exposure, whereas (equilibrium) population abundance is much less dependent and variable (Textbox 2), we only included population abundance in our analysis. An alternative metric that integrates growth rate and abundance is the calculation of the risk equivalent (Nakamaru et al., 2002, 2003): the reduction in population carrying capacity that leads to the same increase of population extinction risk as the chemical exposure at the given concentration. However, we chose to work with

population abundance because it is more intuitive, has a clear link with protection goals, and is therefore more directly amenable for future regulatory applications (Table 1).

Moreover, in biological field monitoring, measures of population abundance are much easier to obtain, and thus more commonly used in assessments of ecological quality (e.g., biological quality elements of the WFD, such as fish population abundance and chlorophyll *a* concentration; Table 1). Thus, modeled population abundance endpoints are easier to relate to common ecological water quality monitoring practice than population growth rates. For

Galic et al., 2010). The scientific community is showing a growing interest in the use of population models for risk assessment because the value of these models is becoming increasingly clear (European Food Safety Authority [EFSA] Panel on Plant Protection Products and Their Residues et al., 2018; Hommen et al., 2010). In addition, the relevance of population models for risk assessment is mentioned in the protection goals of several European guidelines and legislations, such as the Water Framework Directive (WFD); the Regulation on the Evaluation, Authorisation and Restriction of Chemicals (REACH); the plant protection products (PPP) regulation; the biocidal products regulation; and others (Hommen et al., 2010). However, the use of population models in regulations thus far has been limited, especially outside of the ERA of PPP. More has been published on the regulatory use of mechanistic toxicity models on the individual level. In 2006, the Organisation for Economic Co-operation and Development (OECD) described mechanistic modeling as a biology-based approach to analyze toxicity data (OECD, 2006). However, modeling tools were solely considered supplementary tools to conventional statistical analyses (e.g., the dose–response curve). In addition, the EFSA has published a scientific opinion on the use of mechanistic models for regulatory risk assessment of pesticides in aquatic ecosystems (EFSA Panel on Plant Protection Products and Their Residues et al., 2018). Within this opinion, toxicokinetic/toxicodynamic (TKTD) models such as the general unified threshold model for survival (GUTS; Jager et al., 2011) are proposed as tools for the risk assessment of pesticides. Most of the developments have focused on the PPP regulation, where variable exposure patterns and field applications are highly relevant. For metal risk assessment, the focus is more on generic scenarios with continuous exposure under REACH and the WFD. The role of population models in that case is thus more to extrapolate laboratory effects on individuals to the population level to help define predicted no-effect concentrations (PNECs) and environmental quality standards (EQSs).

The use of population models is guided by the protection goals defined in regulations. Table 1 provides an overview of the current population-level protection goals explicitly defined in a few key European chemical risk assessment schemes.

The first aim of the present study was to better understand and estimate how typical ecotoxicity test results translate to population-level effects. We applied population models to toxicity data of four metals: copper (Cu), nickel (Ni), silver (Ag), and zinc (Zn). Based on the REACH dossiers of these metals, a taxonomically diverse set of species was selected for modeling. Criteria for the selected species included the availability of both existing population models and appropriate toxicity data, as well as considering the range of taxonomic groups needed for a valid SSD, according to REACH (European Chemicals Agency, 2008). In total, three primary producers, five invertebrate species, and five fish species were included, with at least one species per trophic level for each metal. Population models were used to extrapolate existing chronic toxicity data for various endpoints to population-level endpoints (e.g., population abundance or biomass). We calculated effect concentrations for population model predictions and compared

these with experimental effect concentrations. In addition, we addressed how different sources of uncertainty contributed to the observed variability across metals and species.

The second aim of the present study was to evaluate the impact of using population models for metal risk assessment. To this end, we present two case studies where population models were used as examples for how they can be incorporated into current risk-assessment practices. First, we evaluated how population model predictions for Cu influenced the SSD to derive safe thresholds. Second, we applied a fish population model to a river system using monitored Zn concentrations to explore how population models can help in the context of the WFD.

EXTRAPOLATION OF METAL EFFECTS TO THE POPULATION LEVEL

Definition of the extrapolation factor

To compare the outcome of chronic toxicity tests with the population models, EC10 values for population endpoints were calculated. We define the ecological modeling-based laboratory to population effect extrapolation factor (for ECOPEX factor) to quantify the differences between laboratory and extrapolated population-level effects.

$$\text{ECOPEX} = \frac{\text{EC10}_{\text{pop}}}{\text{EC10}_{\text{lab}}} = \frac{\text{Predicted EC10 at population level}}{\text{EC10 estimated or calculated in laboratory toxicity test}} \quad (1)$$

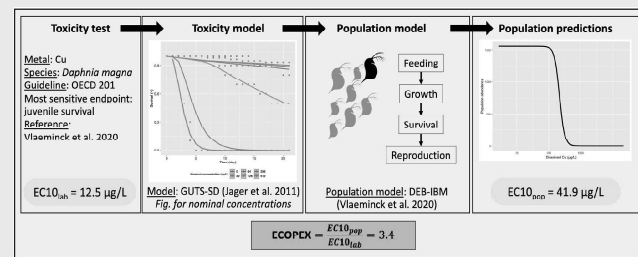
The ECOPEX factor expresses the difference in sensitivity between the laboratory test and the population level (Textbox 1). An ECOPEX factor of 1 means that a 10% effect is observed in laboratory tests and a 10% effect is predicted at the population level at the same exposure concentration. An ECOPEX factor > 1 means that effects at the population level are predicted to be lower than what is measured in a laboratory test; an ECOPEX factor < 1 means that effects at the population level are predicted to be higher than what is measured in a laboratory test (Textbox 1).

Comparison ECOPEX factors between metals and species

Across all metals and species, a wide variety of model types, simulation scenarios (external factors and conditions), original endpoints (e.g., survival, growth, reproduction), and population-level endpoints (e.g., population growth rate or population abundance) was covered. To avoid overly complex integration toward a generic ecological relevancy normalization, some rationalization was needed to retain one ECOPEX factor per metal and species.

First, in terms of population-level endpoints, both modeled population growth rate and population abundance (or biomass) endpoints have been reported across our work (see Supporting Information, S1, for an overview). However, because

TEXTBOX 1: ECOPEX workflow with example



OECD = Organisation for Economic Co-operation and Development; EC10_{lab} = 10% effect concentration in the laboratory; GUTS-SD = general unified threshold model for survival–stochastic death; DEB = dynamic energy budget; IBM = individual-based model; EC10_{pop} = 10% effect concentration in the population; ECOPEX = ecological modeling-based laboratory to population effect extrapolation.

consistency, when both abundance (total number of individuals) and biomass (total mass of individuals) were available population endpoints, abundance was selected for the ECOPEX calculations. In addition, both population density and abundance have been reported in earlier studies. There are two closely related endpoints that either explicitly consider the size of the area or volume where the population lives (density) or make abstraction of this (abundance). For simplicity, we refer to both in the present study as population abundance.

Second, when multiple laboratory EC10 (EC10_{lab}) values were available in the original study for the same individual-level endpoint (e.g., growth) for the same species (e.g., because the same type of toxicity tests had been conducted in multiple test media or under multiple conditions), each EC10_{lab} was separately translated to a population EC10 (EC10_{pop}) to yield as many ECOPEX factors as there were original EC10_{lab} values for that species and individual-level endpoint, and finally the geometric mean ECOPEX factor was calculated for that species and endpoint. Taking the geometric mean of ECOPEX values is equivalent to a common approach in risk assessment for taking the geometric mean when multiple EC10 values are available for the same endpoint of the same species. In some cases, an EC10 was not reported or could not be reliably determined (either for the original study or for the simulated population abundance or for both), and a NOEC was then used instead (see Supporting Information, S1, for details).

Third, when multiple individual-level endpoints were reported in the original study for a species, the ECOPEX factor corresponding with the reported most sensitive individual-level endpoint was retained per species. This makes sense because the hazardous concentration for 5% of species (HCS) based on conventional individual-level ecotoxicity test endpoints is also obtained after selecting the most sensitive endpoint per species.

Fourth, when model predictions had been run under various scenarios, in which the effect of factors like temperature, nutrient or food levels, light intensity, or flow rates were investigated, only the scenario that was closest to the conditions of the original study was retained for our quantitative analysis. The influence of the environmental factors is discussed in more detail further on in the present study.

The ECOPEX factors per metal and species are reported in Table 2 and Figures 1–4. All ECOPEX factors with details on their derivation, the source of the toxicity data, and so on, are reported in the Supporting Information (see Supporting Information, S1). For example, for *Lymnaea stagnalis*, laboratory-derived data on Cu effects on 28-day growth rate and wet weight increase were extrapolated to the population level from two studies and for two different food types (fish flakes and lettuce). First, in this case, a dynamic energy budget individual-based model (DEB-IBM) was calibrated to the laboratory concentration–response data for individual snails (no competition) under the conditions of the laboratory test, including a selection of the most likely physiological mode of action.

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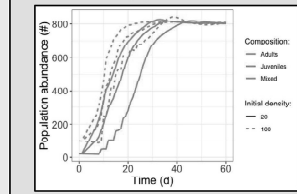
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TEXTBOX 2: Importance of initial simulation settings for population endpoints

Population simulation with *D. magna* DEB-IBM (Vlaeminck et al. 2020) were performed to assess differences in population growth for different compositions of the initial population.

- 20 or 100 individuals
- Adults only (gravid females).
- Juveniles only (neonates of <48h of age).
- Mixed population (50% adults and 50% juveniles).

Daily food addition of 1 mg C per L; temperature is set to 20°C



D. magna = *Daphnia magna*; DEB-IBM = dynamic energy budget individual-based model; Eq. = equilibrium

Effects on population growth rate and equilibrium abundance are assessed. The derived specific growth rates between day 0 and 10.

$$\mu = \frac{\ln(N_t) - \ln(N_0)}{t - t_0} = \frac{\ln(N_{10}) - \ln(N_0)}{10 - 0}$$

$$\mu = \frac{\ln(N_t) - \ln(N_0)}{t - t_0} = \frac{\ln(N_{10}) - \ln(N_0)}{10 - 0}$$

Initial population	Eg. density (H)	Specific growth rate (d ⁻¹)
20 Adults	806	0.28
100 Adults	806	0.31
20 Juveniles	805	0.09
100 Juveniles	805	0.25
20 Mixed	806	0.23
100 Mixed	806	0.29

Based on the predicted population dynamics:

- Initial population size mainly impactful for juveniles.
- Populations with juveniles only takes longer to reach the equilibrium density.
- All populations achieve the same equilibrium population density.

Conclusions

Initial population size and composition are impactful on population growth rate but not for equilibrium population size.

Second, this calibrated DEB-IBM was used to simulate population dynamics at various Cu concentrations under the same conditions but in a population context (with competition for food). Third, the Cu concentration that resulted in 10% reduction of population abundance compared to the control was defined as the EC10_{pop}. When relevant (e.g., ECOPEX factors calculated for different test waters), the range is reported. When no exact ECOPEX factor could be determined (e.g., for the Ag test with *Brachionus calyciflorus*; Table 2), the lowest value was used in figures as the most conservative approach.

Across all modeled species and metals, ECOPEX factors vary from 0.7 to 7.86, with an overall median of 2.8 (Table 2). Effects extrapolated with population models from toxicity tests would thus be either higher or lower than the originally observed effects, although there is a clear tendency for lower effects at the population level. Only in three cases out of 27 was the ECOPEX factor for a species <1, and in none of the cases was the ECOPEX <0.5. In roughly half of the cases (14 out of 27), the ECOPEX factor was >2. All calculated ECOPEX factors for primary producers were >1, suggesting lower effects at the population level than observed in toxicity tests (1.1–7.86, geometric mean = 4.3). For invertebrates (ECOPEX factor 0.7–7.40, geometric mean = 3.0) and fish (ECOPEX factor 0.7–6.0, geometric mean = 1.7), not all population models predicted lower effects at the population level.

Nonetheless, there was a trend for lower effects at the population-level mean, especially for invertebrates. Copper (1.0–3.7), geometric mean = 1.9, Ni (0.7–4.9), geometric mean = 2.7, Zn (0.7–6.0), geometric mean = 1.7) had higher impacts on ECOPEX factors, whereas Ag had noticeably higher ones (1.4–78.6, geometric mean = 11.8). A comparison between lab data and population extrapolations has been performed before, that is, by using Leslie matrices for *Pimephales promelas* for five metals (Wasaki et al., 2010). The approach in the present study differs from our ECOPEX approach because it used calculated population-level threshold concentrations (PTCs) that lead to net zero population growth (i.e., eigenvalue of 1 of the Leslie matrix) instead of the EC10_{pop} for population abundance. Nevertheless, the study also reported a slight tendency toward the population-level endpoints for fish being on average less sensitive than laboratory-derived individual-level endpoints (i.e., geometric mean of 1.1 for their PTC/laboratory-NOEC ratio, range 0.3–3.3).

SOURCES OF UNCERTAINTY AND VARIABILITY

Explaining the observed variability in the ECOPEX factors across metals, across species, and even within species is a

TABLE 2: Ecological modeling-based laboratory to population effect extrapolation factors across metals and species

Group	Species	Model type	Toxicity test endpoints reported		ECOPeS factor						
			Endpoint	Metal	Ag	Cu	Ni	Zn			
Ligae	<i>Dendroicae subspicata</i>	ODE	Population growth rate	Zn	—	—	—	11			
	<i>Raphidocelis subopacata</i>	ODE	Population growth rate	Ag, Ni, Zn	6.9	—	—	1.2			
	<i>Lemna minor</i>	ODE	Total root length	Ag, Cu	78.6	3.7	71 (5.4–9.4)	1.5	—		
	<i>Brachionus calyciflorus</i>	IBM	Population growth rate (juvenile)	Ag, Cu, Ni, Zn	>74	2.8 (1.4–5.4)	—	—	0.8 (0.5–1.1)		
		IBM	Survival (juvenile)	Ag, Cu, Ni, Zn	—	2.5 (0.8–4.2)	—	—	—		
		IBM	Reproduction (juvenile)	Ni	—	—	—	—	—		
		IBM	Survival (juvenile)	Ni	—	—	—	—	—		
	<i>Daphnia longispina</i>	IBM	Reproduction (juvenile)	Ag, Cu, Ni, Zn	4.0 (3.3–5.0)	3.4	7.0 (1.7–15.9)	2.6			
		IBM	Survival (juvenile)	Ag, Cu, Ni	—	—	—	—	1.3		
		IBM	Survival (juvenile)	Ag, Cu, Ni	—	—	—	—	—		
IBM		Individual growth rate (juvenile, wet weight)	Cu, Ni	—	1.6 (0.6–2.7)	9.4 (2.5–38.6)	0.7 (0.5–0.8)				
Ligae ^a	<i>Lymnaea stagnalis</i>	IBM	Individual growth rate (juvenile, wet weight)	Cu, Ni	—	—	—	—	—		
	<i>Aequipecter transtomans</i>	IBM	Survival (juvenile)	Cu, Ni	—	1.0 (0.6–1.3)	—	—	—		
		IBM	Survival (juvenile)	Cu, Ni	—	1.0 (0.3–2.2)	0.7 (0.6–0.8)	3.7 (2.1–4.5)			
		IBM	Survival (juvenile)	Cu	—	1	—	—	—		
		IBM	Survival/growth/reproduction (juvenile/head)	Cu, Ni, Zn	—	—	—	—	—		
	<i>Salvelinus fontinalis</i>	IBM	Survival (juvenile)	Cu	—	—	—	—	—		
		IBM	Larval growth (embryo)	Cu	—	—	—	—	—		
		IBM	Larval survival (embryo)	Cu	—	—	—	—	—		
		IBM	Larval length (embryo)	Cu	—	—	—	—	—		
		IBM	Larval weight (embryo)	Zn	1.5	—	—	—	>6.0 (>4.0–17.6)		
IBM		Larval hatching (embryo)	Zn	—	—	—	—	—			

The study on *Phrynosoma marmorata* and *N.* was not considered because of design and statistical issues with the toxicity test (see Supporting Information S1). The geometric mean ECOPH factor for the minimum and maximum ECOPH factors between parentheses. Brignipus available from the toxicity tests are included, as well as in bold the mean for which this is not the case. In Supporting Information S1, the following abbreviations are used: CQPE = ordinary differential equation; IIM = individual-based model; COPEX = ecological model-based; IIM = individual-based model.

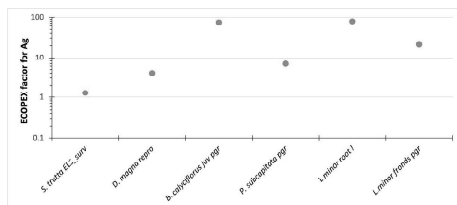


FIGURE 1: Ecological modeling-based laboratory to population effect extrapolation (ECOPEX) factor for Ag for five species. For each species information on life stage and/or endpoint(s) in original study is reported on the x-axis. Orange points indicate which ECOPEX factors were selected per species to determine the ECOPEXgeometric factor. *S. trutta* = *Salmo trutta*; ELS = early life stage; surv = survival; *D. magna* = *Daphnia magna*; repro = reproduction; *B. calyciflorus* = *Brachionus calyciflorus*; juv = juvenile; p_{exp} = (exponential) population growth rate estimated directly from lab test; *P. subcapitata* = *Pseudokirchneriella subcapitata*; *L. minor* = *Lemna minor*; root = root length.

challenge. This may in part be the result of biological variability; that is, the observed variability is the result of real differences between individuals and species. For example, different metals can have different modes of action at the (sub)individual level, which can further differ between species, possibly leading to different ECOPEX factors. However, there is also uncertainty surrounding these ECOPEX factors because of (1) the modeling choices and assumptions made (e.g., type of toxicity model, population endpoint), and (2) the design and reporting of toxicity tests (e.g., chosen endpoint, exposure, data availability).

We identified five main contributors to the variability in ECOPEX factors (across species and metals): (1) uncertainty about the toxicity model, (2) uncertainty about the toxicity

mechanism of the metal, (3) uncertainty caused by test design, (4) impact of environmental factors, and (5) impact of population endpoint chosen. We discuss each of these sources in more detail and provide a few examples. Finally, we draw some general conclusions across all these sources of uncertainties and make recommendations for model development and regulatory applications.

Uncertainty about the toxicity model

Effects observed in laboratory toxicity tests are integrated in population models using toxicity models. Different toxicity

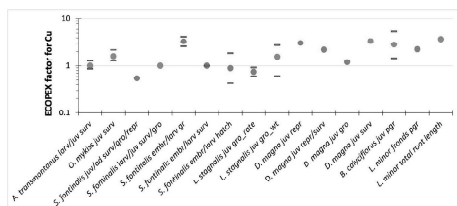


FIGURE 2. Ecological modelling—laboratory to population effect extrapolation (ECOPEX) factor for *Cu* for seven species. For each species information on life stage and/or endpoint(s) in original study is reported on the axis. When more than one life stage is mentioned, it means that the reported test outcome was obtained during an experiment that covered those life stages. When more than one endpoint is mentioned for a given point in the graph, it means that those endpoints all had the same 10% effect concentration (or no-observed-effect concentration) in the original study. The gray horizontal lines represent the 10% effect concentration for each species. The gray horizontal line markers represent the ECOPPEX factors were available for endpoints obtained from more than one original study and/or at least two life stages, the minimum and maximum ECOPPEX factors are depicted with gray horizontal stripe markers. *A. transmontanus* = *Acipenser transmontanus*; larv = larvae; juv = juvenile; surv = survival; *C. oxyrinchus* = *Oncorhynchus mykiss*; *S. fontinalis* = *Salvelinus fontinalis*; ad = adult; gro = growth; rep = reproduction; embr = embryo; *S. alpestris* = *Salvelinus alpestris*; *S. gairdneri* = *Salvelinus gairdneri*; *S. leucomaenis* = *Salvelinus leucomaenis*; *S. malinche* = *Salvelinus malinche*; *B. calyciflorus* = *Brachycaus calyciflorus*; pop = reproductive population growth rate estimated directly from lab test; f. minor = *Lepomis microlophus*.

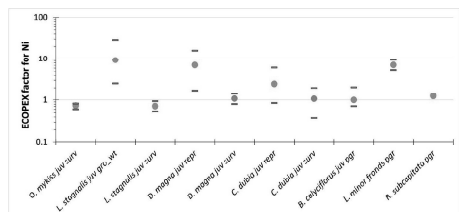


FIGURE 3. Ecological modelling—lab to laboratory to population effect extrapolation (ECEPOX) factor for Ni for seven species. For each species information on life stage and/or endpoint(s) in original study is reported on the x-axis. Orange points indicate which ECEPOX factors were selected per species to determine the ECEPOX geometric factor. In cases where ECEPOX factors were available for endpoints obtained from more than one original study and/or in at least two test media or conditions, the minimum and maximum ECEPOX factors are depicted with gray horizontal stripes. O, mykiss = *Oncorhynchus mykiss*; juv = juvenile; sur = larval; L, signalis = *Lymanea signalis*; gsr, wt = growth based on weight; gsr, vol = growth based on volume; gsr, wt, vol = growth based on weight, volume and weight; gsr, wt, vol, exp = growth based on weight, volume and weight (exponential); pop = population growth rate estimated directly from lab test; L, minor = *Lymanea minor*; P, subcapitata = *Pseudokirchneriella subcapitata*.

models and approaches were used throughout the studies, and although this choice was typically based on the availability of the (raw) toxicity data, it is important to understand how much this choice can potentially affect the ECOPEX factors when interpreting differences in ECOPEX factors. To quantify the impact of this uncertainty, we used the *Daphnia magna* population model and a Cu toxicity data set (Vlaeminck et al., 2020) as a case study. Different sources of uncertainty surrounding the toxicity model were identified and investigated with the *Daphnia magna* model via simulations for their impact on the ECOPEX factors (Textbox 3). Except for the analysis in which the impact of the

dose-response curve (DRC) type was investigated, the log-logistic DRC was chosen for laboratory DRC (DRC_{lab}) and the toxicity DRC (DRC_{tox} ; see Textbox 3 for abbreviations) because this is the most commonly used type. The EC10_{lab} ($12.5 \mu\text{g/L}$) was used as the reference model to calculate the COPEX factor because this is the published value (table 3 in Vlaeminck et al., 2020). In addition, we considered the “exact method,” where we used the population model to estimate the concentration at which exactly a 10% effect at the population level is predicted, the most accurate way to determine the EC10_{pop} . We therefore adopted the exact method as the reference method to

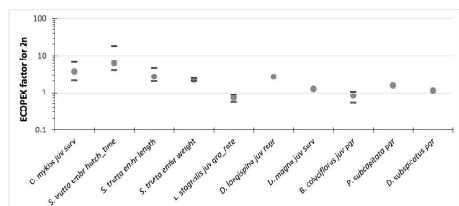
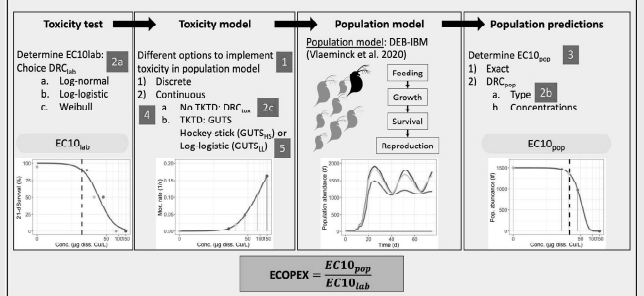


FIGURE 4. Ecological modelling-based laboratory to population effect extrapolation (ECOPLEX) factor for ZoE for eight species. For each species information on life stage (and/or endpoint(s)) in original study is reported on the x-axis. When more than one life stage is mentioned, it means that the reported test outcome was obtained during an experiment that covered those life stages. When more than one endpoint is mentioned for a given point in the graph, it means that those endpoints all had the same 10% effect concentration (or no-observed-effect concentration) in the original studies. The gray horizontal stripes represent the range of ECOPLEX factors calculated from the available data. The minimum and maximum ECOPLEX factors were available for endpoints obtained by more than one original study and/or in at least two test media; the minimum and maximum ECOPLEX factors are depicted with gray horizontal stripe markers. O mykiss = *Oncorhynchus mykiss*; juv = juvenile; surv = survival; S. trutta = *Salmo trutta*; emb = embryo; L stagnalis = *Lymnaea stagnalis*; gro_rate = growth rate; D. longispina = *Daphnia longispina*; repr = reproduction; P. promelas = *Pimephales promelas*; gill_rate = gill respiration rate; gill_rate_estimated = gill respiration rate estimated directly from lab test; D. subspicata = *Pseudokirchneria subspicata*; D. bispunctata = *Daphnia mendotensis* subsp. sp.

TEXTBOX 3: Sources of uncertainty and their impact on the ECOPEX factor investigated using *Daphnia magna* simulations

UNCERTAINTY ECOPEX: SIMULATIONS WITH DAPHNIA MAGNA



Source uncertainty (numbers in scheme above)	Choices	Description	Factor*
1 Extrapolation of toxicity data	Discrete Continuous	With the discrete approach, only the original test concentrations were individually implemented with their observed effects (i.e., the observed percentage effect on growth, reproduction, or survival were implemented at the process level and extrapolated separately). In the continuous approach, a fitted (i.e., calibrated) DRC was used (i.e., a continuous relationship between concentration and effect). The latter has the advantage that population effects at untested concentrations can be simulated. On the other hand, uncertainty associated with the DRC model fitting is then considered in the model, which could lead to additional variability in model predictions. We compared implementing the individual-level responses using the "fitted" observed effects ("discrete extrapolation") or using a fitted DRC ("continuous extrapolation").	1.5
2 Choice DRC type	DRC types: 2a: DRC _{lab} 2b: DRC _{pop} 2c: DRC _{tox}	The choice of the type of DRC could be important, especially at low effect concentrations (e.g., the EC10 range). The log-logistic dose-response model was most often used in our work, even though the Weibull and the log normal (i.e., probit) dose-response models are commonly used as well. Ideally, the best fitting dose-response model should be considered, but was not always done. The type of DRC must be determined both for the individual-level and for the population-level, to determine the EC10 _{lab} (DRC _{lab} 2a) and the EC10 _{pop} (DRC _{pop} 2b), respectively. Additionally, if the toxicity is implemented in the population model using a DRC, the type of the DRC _{tox} and whether it is the same type as the DRC _{lab} can matter (2c). All three DRC types were compared in the <i>D. magna</i> simulations.	2a: 1.3 2b: 1.2 2c: 3.2

(Continued)

TABLE 3: Simulated EC10 values for different combinations of test conditions resulting in simulated EC10_{pop} = 14.4 µg/L. Simulations were done with the *Lyman stagnalis* DEB-IBM using the toxicity model for Cu calibrated in Vlaeminck et al. (2019)

Endpoint	Initial age (days)	Exposure duration (days)	EC10 _{lab} (µg/L)	ECOPEX factor
Dry wt	0	14	13.5	1.1
		28	9.9	1.5
		49	7.6	1.9
		72	10.1	1.4
		14	14	14.0
Shell length	0	14	31.1	0.5
		28	16.7	0.9
		49	16.9	0.9
		72	17.1	0.8
		14	30.8	0.5
Growth rate	0	14	27.8	0.5
		28	22.4	0.6
		49	36.4	0.4
		72	37.4	0.4
		14	30.8	0.5
Reproduction	72	28	20.5	0.7
		49	17.1	0.8
		14	14.0	1.0

EC10_{lab} = 10% effect concentration in the laboratory; EC10_{pop} = 10% effect concentration in the population; DEB-IBM = dynamic energy budget individual-based model; ECOPEX = ecological modeling-based laboratory to population effect extrapolation.

Implementation in a TKTD model is recommended whenever the data needed to calibrate a TKTD model are available.

The difference between discrete and continuous extrapolation (either DRC or GUTS) of the toxicity data was a factor 1.5 in ECOPEX factor. When no toxicity model can be fitted to the toxicity data, this can thus have a considerable impact on the ECOPEX factor. Compared to the discrete extrapolation, the ECOPEX factor values determined

with continuous extrapolation are also closer to the ECOPEX factor of the exact method and were thus considered more accurate. A model that can predict intermediate concentrations (i.e., continuous extrapolation)—provided the model fit is good—is thus recommended over implementing observed effects directly (i.e., discrete extrapolation). The difference between deriving the exact EC10 with the population model compared to using the DRC_{pop} made up to a factor 1.3 variability in the ECOPEX factor. Finally, the choice of the stress-damage relationship in GUTS had a minor impact (factor 1.2 variability).

Uncertainty about the toxicity mechanism

For both lethal (stochastic death [SD]) or individual tolerance (IT) and sublethal (mode of action) effects, uncertainty about the toxicity mechanism will hinder the precise determination of an ECOPEX factor.

In GUTS, different model formulations exist for SD, IT, and the combination of the two. When no survival data over time are available, it is hard to determine whether individuals follow the SD or IT mechanism. The choice for SD or IT has large implications for the population model because under IT, at constant exposure concentrations, individuals surviving the initial exposure are resilient for the rest of their life cycle, whereas under SD they continuously have a probability to die. A good example of this is the population model for *Salvelinus fontinalis* exposed to Cu (Janssen et al., 2024). Because the death mechanism could not be determined, both the SD and IT versions of the GUTS model were extrapolated to the population level. The NOEC for young-of-the-year abundance for SD was 3.4 µg/L compared to a NOEC of 9.5 µg/L for IT.

When implementing sublethal effects in DEB, the toxicant can affect different physiological processes. Which processes are affected and how is described by the physiological model of action (PMA) of a toxicant (Jager, 2020). The PMA of a toxicant is determined based on available toxicity data, sometimes

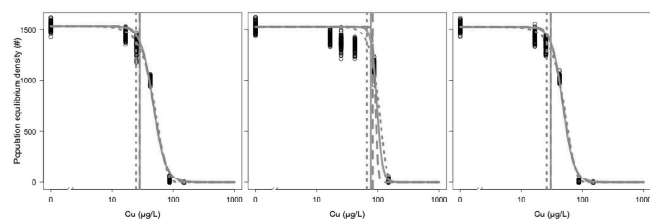


FIGURE 5: Fits of the log-logistic (blue), log-normal (red), and Weibull (purple) type dose-response curves (DRCs) for the population for three different DRC (DRC_{lab}) implementations (left, DRC_{lab}, log-logistic; middle, DRC_{lab}, log-normal; and right, DRC_{lab}, Weibull). Vertical lines show the 10% effect concentration values.

BOX 3: (Continued)

Source uncertainty (numbers in scheme above)	Choices	Description	Factor*
3 Determination EC10 _{pop}	Exact DRC _{pop}	EC10 _{pop} values were determined in different ways. Either the population model was used to estimate the concentration at which exactly 10% effect at the population level is predicted to occur ("Exact") or a DRC was fitted ("DRC _{pop} ") to a set of simulated population endpoints at various concentrations. The exact method was considered the reference method as it exactly determines the EC10 _{pop} . The DRC method has the advantage that EC _x values—once the DRC curve has been determined—can be determined without needing to run the population model. The impact of this choice on the ECOPEX factor was further investigated.	1.3
4 TKTD	GUTS DRC _{tox}	Throughout the population model applications, both toxicity models that did and did not consider TKTD were used. TKTD models such as GUTS (Jager et al., 2011) explicitly account for temporal patterns in toxicity data and are thus more versatile, especially in non-constant exposure conditions. However, they are more complex and have higher data requirements. The choice for no-TKTD toxicity models was often driven by data limitations (i.e., GUTS requires survival measurements over time, which are not always available). The impact of this choice at the population level is not always clear. Thus, the implementation of a TKTD model (GUTS) for lethal effects was compared with a model without TKTD, i.e., a DRC _{tox} model for the mortality rate (de Leender et al., 2009).	1.8
5 Stress-damage relationship	GUTS _{IS} GUTS _{LI}	In TKTD models, the traditional implementation of the stress damage relationship is by using a hockey-stick curve (GUTS _{IS} ; Jager & Ashauer, 2018) but we have also used a log-logistic curve (GUTS _{LI}) for this Vlaeminck et al., 2020). The implication of this assumption was further investigated with simulations.	1.2

EC10_{lab} = 10% effect concentration in the laboratory; DRC_{lab} = laboratory dose-response curve; TKTD = toxicokinetic/toxicodynamic; DRC_{pop} = toxicity dose-response curve; GUTS = general unified threshold model for survival; DEB-IBM = dynamic energy budget individual-based model; EC10_{pop} = 10% effect concentration in the population; DRC_{tox} = population dose-response curve; ECOPEX = ecological modeling-based laboratory to population effect extrapolation.

*Factor: Dividing possible ECOPEX factors (based on the uncertainty source) by the reference ECOPEX factor gives an indication of the impact of the uncertainty source. The factor reported in the present study is the largest possible factor (see Supporting Information, S1, for individual values).

determine EC10_{pop} where possible and expressed deviations from this reference method as factors.

Overall, ECOPEX factors across all simulations varied between 1.2 and 6.3 (see Supporting Information, S2, S3). The largest variability in ECOPEX factor was noted when evaluating the impact of DRC type choice for DRC_{tox} (factor 3.2 difference between ECOPEX factors). The type of DRC_{tox} used to calculate the mortality rate led to clearly higher ECOPEX factors for the log-normal type (5.1–6.3) compared to the log-logistic (2.0–2.3) and Weibull (2.5–2.9) types. The population DRC (DRC_{pop}) fits for the log-normal type were not good, especially for the lowest test concentrations (Figure 5), so we should be careful to not overinterpret the results obtained with this type. This observation highlights the need to thoroughly evaluate DRC fits and not resort to the log-logistic model as a universally applicable model to derive EC10 values. Whether or not the type of DRC was the same for DRC_{lab} and DRC_{pop} did not have that much impact (factor 1.2). For DRC_{lab} (factor 1.3) and DRC_{pop} (factor 1.2), the choice for DRC type was less important because all types resulted in good fits. We also explored

whether it is best to simulate the originally tested concentrations or predict log-equidistant concentrations to obtain an EC10_{pop} as close as possible to the exact method. Across the three types of DRC, simulating log-equidistant concentrations resulted in better correspondence with the EC10_{pop} value derived with the exact method.

The choice of implementing a TKTD toxicity model (DRC_{tox} vs. GUTS) resulted in a factor 1.8 variability. Not implementing a TKTD resulted in prediction of a more sensitive population (EC10_{pop} = 22 µg/L vs. 39 µg/L), but comparison with observations from a population experiment indicated that the TKTD model provides better predictions (9% vs. ~14% deviation from effect at 46.4 µg/L observed in the experiment of Vlaeminck et al., 2020). Compared to statistical methods (DRCs), TKTD models have several advantages: They explicitly consider the time dimension of chemical effects, they can be used to extrapolate beyond the boundaries of tested exposure conditions, and the TKTD parameters can also be interpreted in a physical or biological way (EFSA Panel on Plant Protection Products and Their Residues et al., 2018). Therefore, the

supplemented with existing knowledge. For example, for *Daphnia magna* the PMA was determined for Cu (decreased assimilation of energy; Vlaeminck et al., 2020) and Ni (increased costs for growth; Pereira et al., 2019) based on the available growth and reproduction data. Typically, information on both growth and reproduction is needed to discriminate between the various possible PMAs. In the absence of necessary data, a most probable PMA cannot be determined with the current approach, which can lead to uncertainty about predicted population effects.

The extrapolations for the snail *L. stagnalis* are a good example of this. Typically, for metals, only effects on growth are available as sublethal effects for this species. The lack of reproduction data prevents identification of the most likely PMA; therefore, we have extrapolated multiple possible PMAs to the population level. For example, Ni toxicity data were available for four test waters (Schliek et al., 2010). Differences in extrapolated Ni EC10 values for population abundance between the three evaluated PMAs were between three- and sixfold (see Supporting Information, S1). For Zn, differences in EC10_{pop} values between PMAs ranged from a factor of 1.8 to 2.9 between four test waters (see Supporting Information, S1). Clearly, the correct identification of the PMA can have important implications for the ECOPEX factor, and the uncertainty is largely dependent on toxicity data of sufficient quality and quantity.

Uncertainty caused by the design of the toxicity test

Next to the uncertainty caused by the modeling of toxicity, there is also uncertainty caused by the variability in toxicity test design, for example, endpoint measured, exposure duration, age of individuals tested (not considering differences in environmental factors such as temperature, food, test medium, and water physicochemistry—these are discussed in the section Impact of environmental factors on the ECOPEX factor). For algae, the use of flow-through or batch test systems was found to influence the ECOPEX factor. In the case of Zn, simulations with a flow-through system were more sensitive compared to batch test systems for *Desmodesmus subspicatus* but this was not observed for *Raphidocelis subcapitata* (Hansel et al., 2021a, 2021b). The hypothesis is that the maximum growth rate of *Desmodesmus subspicatus* was close to the dilution factor in the flow-through system, which led to higher sensitivity in the test and thus a lower ECOPEX factor.

The design of toxicity tests is often guided by practical decisions, for example, the focus on early life stages or short exposure durations; and these design choices can lead to different EC10_{lab} values. This is mostly relevant for species where testing the whole life cycle is a challenge. For example, for *L. stagnalis* and the four fish species modeled, the available data from the toxicity tests were vastly different (Supporting Information, S5).

Differences in test design (choices of life stage, initial size/age, endpoint, exposure duration) can result in different EC10_{lab} values, even though the toxic effect of the metal is identical in

the studies. Hypothetically, in such instances, even with identical population effects (i.e., the same predicted EC10_{pop}), different ECOPEX factors could be derived because of the differences in EC10_{lab} values. To better understand the potential impact of test conditions, additional simulations with the *L. stagnalis* DEB-IBM were performed, using the implementation and parameterization of Cu toxicity as in Vlaeminck et al. (2019; PMA = decreased assimilation of energy). We investigated how different test design choices can affect the EC10_{pop}, even if the modeled Cu toxicity at the population level (EC10_{pop} = 14.4 µg/L) is identical. Based on observed differences in the design of toxicity tests for Cu, Ni, and Zn (Supporting Information, Table S5), a different initial age (0, 14, 21, and 72 days old), exposure duration (14, 28, 49, and 98 days), and reported endpoint (biomass, shell length, growth rate, and reproduction) were simulated (Table 3). Simulated EC10_{lab} values varied between 7.6 µg/L (dry wt of age 0 d) and 41.6 µg/L (shell length of age 72 days) or a factor 5.5 difference. The ECOPEX factors were between 0.3 and 1.9. The choice of endpoint had the largest impact, leading to a factor difference in EC10_{lab} values between 1.7 (age 14 days, dry wt vs. shell length) and 4.1 (age 0 days, dry wt vs. shell length). Dry weight was in general the most sensitive endpoint, whereas shell length was the least sensitive. At age 0 days, individuals were always the most sensitive, with a factor difference between 1.3 (shell length, age 72 days vs. age 0 days) and 1.9 (dry wt, age 72 days vs. age 0 days). Exposure duration led to differences between a factor 1 (age 72 days, reproduction) and 1.9 (age 72 days, shell length). Even though the Cu toxicity at the population level was identical, EC10_{lab} and consequently ECOPEX factors were very different depending on the test design.

Impact of environmental factors on the ECOPEX factor

In a small number of the performed modeling studies, the direct impact of environmental factors on population predictions was investigated. More specifically, the effect of food quantity and quality, the temperature, and the physicochemistry of the test water on population-level sensitivity were investigated.

In the case of *L. stagnalis* and Cu, data were available from toxicity tests that had been performed with different food sources, fed ad libitum (Vlaeminck et al., 2019): fish flakes, lettuce, and a mixed food. The individual-level sensitivity of *L. stagnalis* depended on the food source. Food effects were implemented by calibrating the population model per food source to the control treatment. In addition, the toxicity model for sublethal effects was parameterized per food source (i.e., different toxicity parameters per food source). Population predictions resulted in ECOPEX factors of 0.6 (fish flakes), 2.2 (lettuce), and 2.7 (mixed food sources).

The effect of temperature on toxicity was experimentally investigated for *Daphnia magna* and Ni, both in toxicity tests (Pereira et al., 2018) and at the population level (Pereira et al., 2019). Temperature had a clear effect on the outcome of the toxicity test: EC10_{lab} values were <4, 13.5, and 88.1 µg/L for

15 °C, 20 °C, and 25 °C, respectively. Predicted population EC10 values were higher and showed much less difference between temperatures (EC10_{pop} values 43.7–171 µg/L at 15 °C–25 °C) and were in line with the observed lack of effect on populations at 12 and 100 µg/L. Consequently, ECOPEX factors varied between 1.7 and 15.9.

For metals, the physicochemical of the water has a large impact on the bioavailability of the metal and thus the toxicity (Paquin et al., 2002). Often, toxicity tests with different test waters were performed to account for this uncertainty, and thus multiple test waters could be extrapolated with population models. Extrapolation of different test waters resulted in a range of ECOPEX factors, for example, for *B. calyciflorus* and Cu (1.4–5.4), Ni (0.7–2.1), and Zn (0.5–1.1); Vlaene et al., 2024) and for *L. stagnalis* and Ni (2.5–28.6) and Zn (0.5–0.8); Vlaeminck et al., 2019; Weighman et al., 2021). In the absence of more knowledge on bioavailability effects at the population level, the geometric mean EC10_{pop} across test waters was taken, which is often done if at least four values for the same species are available (European Chemicals Agency, 2008). Bioavailability models can help reduce the uncertainty. In the case of *Oncorhynchus mykiss* with Cu and Zn, a bioavailability model (biotic ligand model [BLM]) was developed for one key parameter (median threshold for damage m₀) of the GUTS toxicity model for survival (Janssen et al., 2021). The GUTS-BLM was able to predict (independent) 30-day median lethal concentration values within a twofold error, which is typically used as a quality criterion for validity of BLMs (De Schampelaere & Janssen, 2002).

Understanding how and why environmental factors affect the ECOPEX factor requires looking into environmental effects on both the intrinsic sensitivity of individuals (the denominator in Equation 1) and the population-level sensitivity (the numerator). The examples given above only extrapolate the toxicity tests to the population level. The model was calibrated to each individual toxicity test, keeping the environmental factors constant between laboratory test and population (except the context of interacting individuals within a population). However, to understand the mechanistic basis for how environmental factors alter the ECOPEX factor, a more in-depth analysis would be required, for example, by using the toxicity model calibrated for one level of an environmental factor (at the individual level) to extrapolate to multiple levels of that factor at the population level. Using the temperature effect on *Daphnia magna* Ni toxicity as an example, the toxicity model calibrated for Ni effects at 20 °C could be used to predict population effects at 15 °C, 20 °C, and 25 °C. Fully differentiating between the effects of an environmental factor on the individual sensitivity and on the population sensitivity was outside the scope of our study, and it is therefore difficult to draw conclusions on how and why environmental factors alter the ECOPEX factor. However, considering the reported variation above, this is an important avenue of further research.

Impact of the chosen population endpoint

Population models allow us to investigate multiple aspects of the population, for example, abundance, biomass, growth

rate, size distribution, or age structure. For simplicity, and because this is often of most regulatory relevance, we have chosen to calculate the ECOPEX factor based on population abundance when multiple endpoints are available. However, there is heterogeneity in the ECOPEX factors because sometimes only population biomass is available (e.g., for algae and plants, for *L. stagnalis* and Zn). This choice does not impact the ECOPEX factors substantially (<30% difference; Supporting Information, S1) in the absence of other uncertainties (e.g., unclear PMoA) but can be significant if, for instance, the PMoA is unclear. For *L. stagnalis*, for example, the PMoA of Ni is uncertain (Weighman et al., 2021). Three PMoAs were extrapolated to the population level. For the PMoA reduced assimilation of energy, the ECOPEX factors for population biomass and abundance are very similar (e.g., EC10_{pop} values of 2.83 and 2.77 µg/L for Calappa test water, respectively). However, for the PMoA increased costs of maintenance, the EC10_{pop} values between population biomass and abundance were very different, for example, a factor 118 different for the Zolner test water (0.57 vs. 67.15 µg/L for biomass and abundance, respectively).

The regulatory context can likely guide the choice of the population endpoint. For the WFD, for example, abundance and age structure are explicitly mentioned as endpoints for fish populations (European Commission, 2000). For other regulations, the protection goal is less clear (see European Chemicals Agency, 2008), and the choice of the population endpoint will impact the ECOPEX factor.

The impact of uncertainty on the ECOPEX factor

Table 4 summarizes the influence of different sources of uncertainty about the ECOPEX factor, based on the case studies (see Supporting Information, S1) as well as the additional simulations performed in the present study. They are only indicative for the potential impact of these uncertainties because they are dependent on the chosen examples; for example, simulations for the impact of test design would yield other results for *Daphnia magna* compared to *L. stagnalis*. The variation in ECOPEX factors because of uncertainty (between 1 and 12) is considerable compared to the variation in ECOPEX factors from the case studies (between 0.7 and 78.6; Table 2).

TABLE 4: Influence of the different sources of uncertainty on the ECOPEX factor, based on the examples given in the present study

Source of uncertainty	Factor variability ECOPEX
Uncertainty about the toxicity model	1.2–3.2
Uncertainty about the toxicity mechanism of the metal	1.4–6.0
Impact of test design	1–5.5
Impact of environmental factors	1.2–12
Impact of population endpoint chosen	1–13 (118)

The factor variability is based on the differences in ECOPEX factors caused by the uncertainties per case study (see Supporting Information, S1). The range of factor variabilities represents differences between case studies. ECOPEX = ecological modeling-based laboratory to population effect extrapolation.

However, not all sources of uncertainty will apply simultaneously, and much of this uncertainty is also present in traditional ERA, for example, variation between toxicity tests. Although there is uncertainty, ECOPEX factors give a good indication of metal effects extrapolated to the population level.

The lack of suitable toxicity data for model calibration had a big impact. This prevented the determination of the PMoA (uncertainty about toxicity mechanism) and guided the choices for the toxicity models implemented. Especially in the case of DEBtox, data on the effects on both growth and reproduction are essential. However, typically only one of these is available, or when available, they are often not measured simultaneously. Such data limitations restrict the ability to determine a most plausible PMoA because oftentimes multiple PMoAs can explain observed toxicity responses for a single endpoint. In such cases we extrapolated all potential PMoAs to the population level. This resulted in a wide range of ECOPEX factors for the same study. PMoA uncertainty led to a difference of up to a factor 6 in ECOPEX factors. The uncertainty about the PMoA also interacted with other sources of uncertainty. For example, because no PMoA could be decided between the test waters for *L. stagnalis* exposed to Ni (Weighman et al., 2021), the impact of water chemistry on the ECOPEX uncertainty is hard to assess. The maximum factor 12 variability in ECOPEX factor caused by environmental factors is thus likely also partly a result of the inability to determine the most likely PMoA. Not only does the inability to identify the PMoA lead to uncertainty about the extrapolations, but it also prevents a deeper understanding of the mechanisms behind the toxicity.

In addition, the (un)availability of toxicity data guided the type of toxicity model implemented. We explored the implications of these modeling choices for lethal effects with *Daphnia magna*, which led up to a difference of 3.2 in ECOPEX factor. This applies as well for sublethal effects, for example, choice of model complexity (to include TKTD), but was not further explored. The optimal data set consists of effects over time and with clear effects. Ideally, the data as recommended in the EFSA scientific opinion on TKTD modeling are available (EFSA Panel on Plant Protection Products and Their Residues et al., 2018), but even the recommendations for a good DRC data set are often not met; for example, no partial effect or 100% effect concentrations were available. The OECD guideline for chronic *Daphnia* tests (OECD, 2012), for example, only requires reporting of survival, time to first reproduction, and reproduction over 21 days. Growth is mentioned but not required by the guideline. Ideally, for future model calibrations, test guidelines should be updated to maximize their potential as good data sources, for example, by reporting endpoints over time and including both growth and reproduction. The increased regulatory attention for model applications creates opportunities for modelers to be more proactive, for example, by getting more involved in the design of testing guidelines.

A general trend across species and metals is that the ECOPEX factor decreased the range of available EC10_{pop} values, especially when considering environmental factors (e.g., water chemistry, temperature). The water chemistry determines the bioavailability of metals, which is an important

determinant of metal toxicity. Calculation tools such as the BLM have been developed to correct for differences in water chemistry (De Schampelaere & Janssen, 2002). For *O. mykiss*, water chemistry corrections were combined with population modeling, and the original variability in EC10_{pop} was reduced from a factor of almost 40 to a factor 1.7 difference between ECOPEX factors (Janssen et al., 2021). When multiple EC10_{pop} values are available, extrapolations with population models can thus help to reduce the differences between studies.

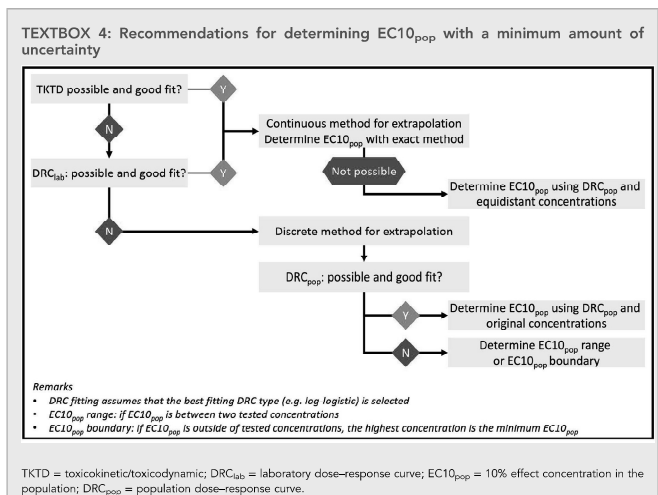
Some of the sources of uncertainty are related to the design of the toxicity tests (e.g., tested life stage, timing and duration of exposure, monitored endpoint). Toxicity tests are typically designed to identify toxicant effects on individuals in a practical way. This is not necessarily the test design that leads to the optimal information to determine effects at the population level. In the case of trout species, for example, effects on early-life stages are less important compared to effects on older individuals because of the already high early-life stage mortality (Janssen et al., 2021), yet this is the most tested life stage. For *Daphnia magna*, it was found that lethal effects can be the dominant driver at the population level, even though reproduction is a more sensitive endpoint (Pereira et al., 2019; Vlaeminck et al., 2020). Ideally, information on different endpoints in the whole life cycle is available, but this is often not practical. Population models provide an opportunity to help design the toxicity test and best determine effects at the population level: Different design options (e.g., life stage, endpoint, exposure) can be explored using population models. This could inform new and existing testing guidelines, especially considering new emerging topics such as endocrine disruptors which can also be modeled (Crane et al., 2019; Forbes et al., 2019).

In general, the choice of population endpoint had a small impact on the ECOPEX factor (typically <30%; see above, Impact of the chosen population endpoint). However, in the case of *L. stagnalis*, up to a factor 118 difference in ECOPEX factors was observed between population abundance and biomass because the PMoA was unclear. Differences in effects between population endpoints do occur. This highlights the need for clear, specific population-level protection goals so that the choice of the most appropriate population endpoint is clear.

Recommendations for future applications

Based on the experiences gained in our work, we propose the following recommendations for minimizing uncertainty and maximizing the potential of population models in ecological risk assessment of metals, and compounds in general.

- The design of toxicity tests should include multiple endpoints, covering at least growth, reproduction, and survival. Ideally, the effects are reported at different points in time.
- Recommendations and guidelines for the generation of good toxicity data should be followed.
- TKTD models: EFSA scientific opinion on TKTD models (EFSA Panel on Plant Protection Products and Their Residues et al., 2018).



- DRC, for example, REACH Guidance R.10 (European Chemicals Agency, 2008).
- Raw data of toxicity tests must always be included in official reports and publications.
- Population models can be used to explore the relevance of test designs for the population level. This can help decide on practical aspects of the test, for example, what life stage to focus on or which endpoints to measure.
- Based on these simulations, to avoid uncertainty as much as possible for future work, we propose recommendations for determining EC10_{pop} and ECOPEX factor as outlined in Textbox 4.

VALIDATION OF POPULATION MODELS

To improve the regulatory use of models, risk assessors need to be convinced of their accuracy and potential. Therefore, validation of the models is critical, that is, demonstrating that the models work as well for independent data not used in the calibration (Forbes et al., 2009; Grimm & Martin, 2013).

However, the availability of good data on toxicity to populations in the field is a challenge. Semifield studies are one of the available alternatives. However, the interaction with other species in the environment can be challenging for population models. Indeed, the population models for *Daphnia magna* and *L. stagnalis* were applied to mesocosm data for Ni (Weighman et al., 2021). These mesocosm studies were not designed to validate population models, and certain model assumptions were therefore required, for example, for the interspecific competition and predation pressure. Although the population models were able to predict some of the patterns in *Daphnia magna* and *L. stagnalis* population dynamics and the effects of Ni, there is a large uncertainty associated with these validations because of the many model assumptions. Another alternative to validate the models is using population experiments in the laboratory. For a few case studies, we were able to do this. The population effects of Cu and Zn on *Daphnia magna* were accurately predicted using the toxicity models calibrated on individual-level data (Vlaeminck et al., 2020). Similarly, for *B. calyciflorus*, we were

able to predict population effects of Cu (Vlaene et al., 2024). This is strong evidence that the developed population models perform as intended. However, for species with longer life cycles such as snails and fish, such population data are absent. A last alternative for such species, which is often the only feasible one, is to validate the population model separately from the toxicity model. This was done in the case of *Salmo trutta* and Zn. The INSTREAM population model was validated for the river Belaguna in northern Spain (Ayllón et al., 2016). The toxicity model was calibrated and implemented in the validated population model. Although not a full validation, this approach does allow us to validate submodels of the whole modeling system. Following this, model acceptance would be helped by a clear and transparent description of the conceptual model, such as the Pop-GUIDE framework (Raimondo et al., 2021) and standardized (i.e., recognizable) conceptual model diagrams (Forbes, 2024).

Risk assessors are usually not trained in modeling and/or in assessing the appropriateness and validity of a model. Efforts have been and are being made to promote the use of ecological models for ERA, for example, the European Union-funded CREAM project (Grimm et al., 2009). Similarly, good modeling practices, such as the ODD (overview, design concepts, and details; Grimm et al., 2020) protocol and TRACE (transparent and comprehensive ecological modeling; Ayllón et al., 2020), have been developed to promote their use (Grimm et al., 2014; Schmolke et al., 2010). Finally, the Society of Environmental Toxicology and Chemistry working group Model Acceptability and Scenario Development is working on acceptance for effect models.

POTENTIAL APPLICATION 1: ECOPEX AND SSD

Species sensitivity distributions are used in the REACH risk assessment dossiers of all four metals considered in the present study. The HC5 of this SSD (here referred to as laboratory HC5 [HC5_{lab}]) serves as a basis for the PNEC value (European Chemicals Agency, 2008). If we want to obtain a PNEC value that is more based on population endpoints, we propose that it makes sense to include the SSD toxicity data that have been extrapolated to the population level (using ECOPEX factors). We chose Cu as an example because the PMoA of Cu was clear for all investigated species, thus avoiding large uncertainty of the ECOPEX factor within one species, and because the toxicity data of the SSD used under REACH are publicly available (see Supporting Information, S4). In total, 27 species with toxicity data (i.e., 27 EC10_{pop} values) are included in the Cu SSD.

Out of six distributions evaluated (log-normal, log-logistic, Weibull, gamma, exponential, and Gumbel), the Gumbel distribution was the best-fitting based on the Anderson-Darling goodness-of-fit test and used to determine the population HC5 (HC5_{pop}). Because no ECOPEX factors are available for all species, we explored different potential approaches and evaluated their impact on the HC5_{pop} (Table 5). The ECOPEX factor values were either applied to all species in the SSD (Approaches 1.1, 1.2, and 1.3) or only to the species with a population model available (n = 5; Approaches 2.1 and 2.2). Because ECOPEX factor values were not available for all species, we applied either the geometric mean ECOPEX factor (1.9 for Cu) to the original HC5_{lab} (Approaches 1.1 and 1.2) or

TABLE 5: Hazardous concentration for 5% of species in a population values (and 95% confidence intervals based on bootstrapping) determined for different possible approaches to apply an ECOPEX correction to the species sensitivity distribution (log normal distribution). The original toxicity data were normalized for the river otter (ecoregion scenario: medium rivers with flow rate of ±200,000 m³/day; United Kingdom)

Approach	HC5 _{pop} (µg/L)	Factor difference
1. For the original SSD with all species (n = 27).	13.7 (9.7–19.4)	–
1.1. Apply geometric ECOPEX factor (1.9) to original HC5 _{lab} .	25.9 (18.5–36.9)	1.9
1.2. Calculate EC10 _{pop} values by applying species-specific ECOPEX factor to species with population model, and apply group-specific (e.g., <i>daphniids</i> , fish) ECOPEX factor for those species without population model. If no ECOPEX factor is available for the group, the geometric ECOPEX factor is used. Then fit SSD to EC10 _{pop} values, and determine HC5 _{pop} .	20.7 (14.4–29.4)	1.5
1.3. Bootstrapping approach: Per iteration (n = 1000), a random ECOPEX factor was sampled from the geometric ECOPEX factors for the group the species belongs to. If no ECOPEX factor was available for the group, the ECOPEX factor was sampled from all available ECOPEX factors. For species with a population model, the available ECOPEX was used. HC5 _{pop} values were calculated for each iteration.	20.2 (18.0–22.3)	1.5
2. SSD for only those species with population model (n = 7).	7.72 (2.68–17.2)	–
2.1. Apply geometric ECOPEX factor (1.9) to original HC5.	14.7 (5.09–32.6)	1.9
2.2. Apply ECOPEX factor to individual species to determine EC10 _{pop} , then fit SSD.	15.7 (6.08–33.3)	2.0

ECOPEX = ecological modeling-based laboratory to population effect extrapolation; HC5_{pop} = hazardous concentration for 5% of species in a population; SSD = species sensitivity distribution; HC5_{lab} = hazardous concentration for 5% of species in a laboratory; EC10_{pop} = 10% effect concentration in the population.

used a bootstrapping approach (Approach 1.3). In the bootstrapping approach, if no species-specific ECOPEX factor was available, ECOPEX factors were sampled from available ECOPEX factors (Table 2), either from ECOPEX factors from the same group if available (e.g., daphnids, fish, gastropods) or from all possible ECOPEX factors. This sampling was iterated 1000 times, and the median HCS_{pop} was calculated from all available HCS_{pop} values based on best-fitting distribution per iteration).

Depending on how the ECOPEX factor was applied to the SSD, the HCS_{pop} was between 1.5 and 2.0 higher than the original HCS_{sub} . The easiest approach is to apply the geometric mean of available ECOPEX factors (1.9; see Approaches 1.1 and 2.1) to the HCS_{sub} , because this ignores possible effects on the fitting of the SSD, the effect of ECOPEX correction on the HCS_{sub} for only modeled species could be considered more representative (Approach 2.2). With this approach, the highest change in HCS_{pop} value was observed: HCS_{pop} is 2.0 times higher than HCS_{sub} . However, because of the low number of data points, the uncertainty on the HCS is higher. In addition, the investigated species (those with population models available) were selected because of their sensitivity to Cu; that is, they are on the low end of the SSD, and an ECOPEX correction of 2.0 could thus not be representative. Assuming that the calculated ECOPEX factors are representative for model-based extrapolations to the population level in general, the bootstrapping approach (Approach 1.3; HCS_{pop} 1.5 times higher than HCS_{sub}) is more realistic.

A comparable exercise was done before for Cu and Zn (Iwasaki et al., 2013; Kamo & Naito, 2008). These studies applied the SSD on PTC (i.e., the concentration above which extinction is predicted), calculated using Leslie population matrices. Comparison with the traditional SSD approach (6.1 $\mu\text{g/L}$) showed similar HCS values for Cu when using the PTC (6.8 $\mu\text{g/L}$), whereas for Zn the HCS based on the PTC (107 $\mu\text{g/L}$) was a factor 7 higher than the traditional approach (14.7 $\mu\text{g/L}$). Direct comparison between their study and ours is not possible because the underlying assumptions are different. The PTC is the concentration above which a population is predicted to go extinct (Iwasaki et al., 2013), whereas with the ECOPEX, we consider the concentration that leads to a 10% reduction in population equilibrium abundance. If an HCS is derived from the SSD, the former approach expresses the concentration when 5% of the species are lost, whereas the latter approach expresses the concentration where 5% of the species experience a 10% or higher reduction of population abundance.

POTENTIAL APPLICATION 2: WFD

Fish population endpoints are explicitly mentioned in the WFD, such as species abundance and age structure (Table 1). Fish population models can provide this information and are therefore useful to assess the impact of metal concentrations in water bodies. In this section, we explore how a population model (INSTREAM-gen) can be used to predict the effects of measured Zn concentrations on rainbow trout (*O. mykiss*) populations.

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From the Waterbase monitoring database, curated by the European Environment Agency, a time series of Zn concentrations was extracted to use as the exposure concentration in the model simulation. The INSTREAM-gen model is parameterized for a 305 m reach in the Belaguer River in northern Spain, but no Zn measurements for this river were readily available. Because parameterization of the model for a new river system was outside the scope of the present study, we applied monitored concentrations of a different river as a proof-of-principle. Because no usable data from Spain are available in the Waterbase, the Serio River (northern Italy) was selected for the case study. This is also a mountain river; measurements of both Zn and water chemistry are available with good frequency throughout the year, and Zn concentrations are high enough that effects can be expected. For the Montedine measuring station, seven dissolved Zn measurements throughout 2020 were available (between 50 and 500 $\mu\text{g/L}$). Measured Zn concentrations and local water chemistry (i.e., dissolved organic carbon, pH, and water hardness) were put into Windermere Humic Acids Model 7 software (Tipping et al., 2011) to estimate free Zn^{2+} ion activity (standard assumptions; see Janssen et al., 2021). Linear interpolation was used to derive daily concentration measurements as input for the model. The time series was then further extended from a 1-year concentration profile to a 10-year concentration profile to assess the impact over a more extended period of time.

INSTREAM-gen is an individual-based model, implemented in NetLogo, that predicts population dynamics of trout species. Toxicity parameters for Zn based on Zn free ion activity, calibrated for *O. mykiss* by Janssen et al. (2021), were used. The parameter set for the Mg-B test water was selected (dominant rate constant $k_d = 0.4 \text{ day}^{-1}$, threshold distribution $m_{th} = 1389 \text{ nM}$, and background mortality $b_{th} = 2.34 \times 10^{-7} \text{ day}^{-1}$) because this set has the lowest median of the threshold distribution (m_{th}) for effects (i.e., it is the most conservative set of parameters). Importantly, following the toxicity data available, only Zn effects on juveniles (<1 year) are implemented in the IBM. For further details on the implementation, we refer to Janssen et al. (2021). Two scenarios were simulated: a control population (unexposed) and an exposed population (with measured Zn concentrations). For both scenarios 100 iterations were simulated. To evaluate potential effects of the Zn exposure, we looked at two relevant population-level endpoints, following the WFD: total population abundance and age structure. We looked at one time point in late summer (day 240, late August) to not overcomplicate the analysis: At this time point, most reproduction has occurred, and abundances give a good overview of expected effects. To account for the stochasticity in the simulations, the probability distributions of 100 simulations with and without exposure were compared. We were particularly interested in simulations that were outside the expected range, which we defined as $P10_{>}$, the percentage of simulations where population abundance is below the 10th percentile of the abundance in control simulations. A $P10_{>}$ of 10%, for example, total population abundance indicates that Zn exposure leads to similar populations compared to the control simulations. A value >10%, indicates that simulations with Zn exposure tend to be impacted; for example, $P10_{>}$ 50%

presented two possible applications: improving available SSD extrapolations and retrospective risk assessment in context of the WFD. Such applications demonstrate that population modeling can already play a role in current ERA practices but requires further promotion for general acceptance. Model communication and transparency are essential, and there are already many good initiatives to promote this, such as PopGUIDE and TRACE. Further validation of population modeling results is critical in this context and should be a high priority.

Supporting Information—The Supporting Information is available on the Wiley Online Library at <https://doi.org/10.1002/etc.5966>.

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Data Availability Statement—All data, analyses, and simulations performed are available in Supporting Information. For the data, code, and simulations of referenced studies, we refer to original publications or reports.

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TABLE 6: Percentage of Zn exposure simulations ($n = 100$) in which population abundance is below the 10th percentile of the abundance in control simulations for total trout abundance and abundance of individual age classes. Abundances were counted at the end of summer (day 240).

Year	Abundance total	Abundance age 0	Abundance age 1	Abundance age 2	Abundance age 3+
Year 0	22	9	98	9	13
Year 1	62	21	98	98	7
Year 2	52	10	98	98	60
Year 3	30	6	98	98	94
Year 4	18	2	98	95	86
Year 5	44	16	98	95	89
Year 6	30	18	98	97	90
Year 7	4	0	98	97	74
Year 8	29	20	98	95	66
Year 9	29	2	98	61	70
Year 10	28	21	98	95	46

indicates that in 50% of simulations with Zn exposure the population abundance is below the 10th percentile of the population abundance in the control.

Effects on total abundance varied through time (Table 6): $P10_{>}$ varied between 4% (Year 7) and 62% (Year 1). There was only 1 year (Year 7) where $P10_{>}$ was <10%; in the other years $P10_{>}$ was >18%. This indicates that effects on fish abundance were expected for almost all years at Zn concentrations measured in the Serio River. Differences in Zn effects over

10 years indicate that other environmental factors (especially climate) affected the magnitude of Zn effects.

Looking at effects on different age classes (Table 6 and Figure 6), effects were most pronounced for fish of 1 year old and older. $P10_{>}$ was often >90%, especially for 1-year-old individuals. For older age classes, abundances overlapped more (Figure 6). Because effects were only implemented for juveniles, effects on older individuals were still absent the first year (Year 0). The more limited effects on total abundance

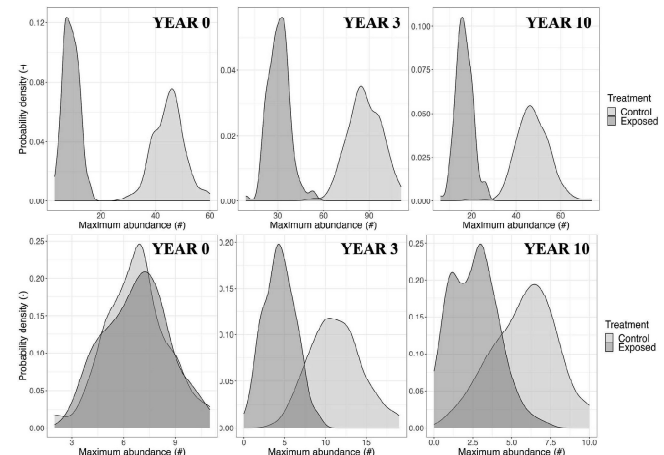


FIGURE 6: Probability distributions for Age 1 (top) and Age 3+ (bottom) trouts (100 simulations), reflecting the stochasticity in the model. The abundance in late summer (day 240) is depicted for 3 years: Year 0 (1994), Year 3 (1997), and Year 10 (2004).

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compared to effects on, for example, Age 1 abundance was because Age 0 individuals dominated the total abundance (>90% contribution to total population abundance), and effects were less severe for this age class. Even though Age 0 individuals were the only sensitive stage for Zn, effects on this age class were limited because of density-dependent compensation mechanisms at the population level. There was a very high (natural) mortality for juveniles, for example, because of competition for food; and effects of Zn reduced the contribution of such density-dependent processes, which compensated for the direct effects of Zn.

Overall, the population model indicates effects of Zn at the population level, not unexpected given the measurements above the EQS of Zn (10.9 $\mu\text{g/L}$). But it also shows the nuance in predictions, where a distinction is possible between age classes and over the years, considering other environmental variables. In addition, adaptation to high concentrations of Zn has been shown in wild brown trout populations in England (Uren Webster et al., 2013) but is not accounted for in the current model. The application presented offers a potential use of population models in retrospective risk assessment in the context of the WFD: What effects can we expect at measured concentrations? Are measured metals possibly driving effects in the field? This population model can also be used in a more prospective risk assessment: At the EQS, do we expect effects? The approach also allows us to account for bioavailability effects and for fluctuations in concentrations over time and (although not explored in the present study) in space. Coupling with environmental fate models/scenarios further increases the usefulness. For metals like Cu (as active substance) or Ag (as bioactive agent) detailed exposure scenarios need to be developed as part of registration dossiers. Predicted concentrations can be used as input to the population model, allowing a realistic prediction of effects in the field.

CONCLUSIONS

With the present study, we have demonstrated that population models can be used to further the ecological relevance of metal ERA in regulatory applications. The ECOPEX factor was introduced as a measure to quantify the difference between toxicity observed at the individual level in laboratory tests to ecologically more relevant population-level toxicity predicted using population models. Overall, predicted population-level toxicity was less sensitive than laboratory toxicity data but not always. This approach is expected to give a more ecologically realistic view on ecotoxicological effects on species. In the present study, we have identified sources of uncertainty that make it difficult to compare ECOPEX factors between species and metals. However, it is important to mention that many of these uncertainties also apply to existing toxicity data, for example, differences in test conditions. Further work on these uncertainties is absolutely required but does not prohibit population modeling from being applied in practice. The ECOPEX factor helps to quantify the potential regulatory impact of population model extrapolations of available laboratory toxicity data to the population level. We have

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